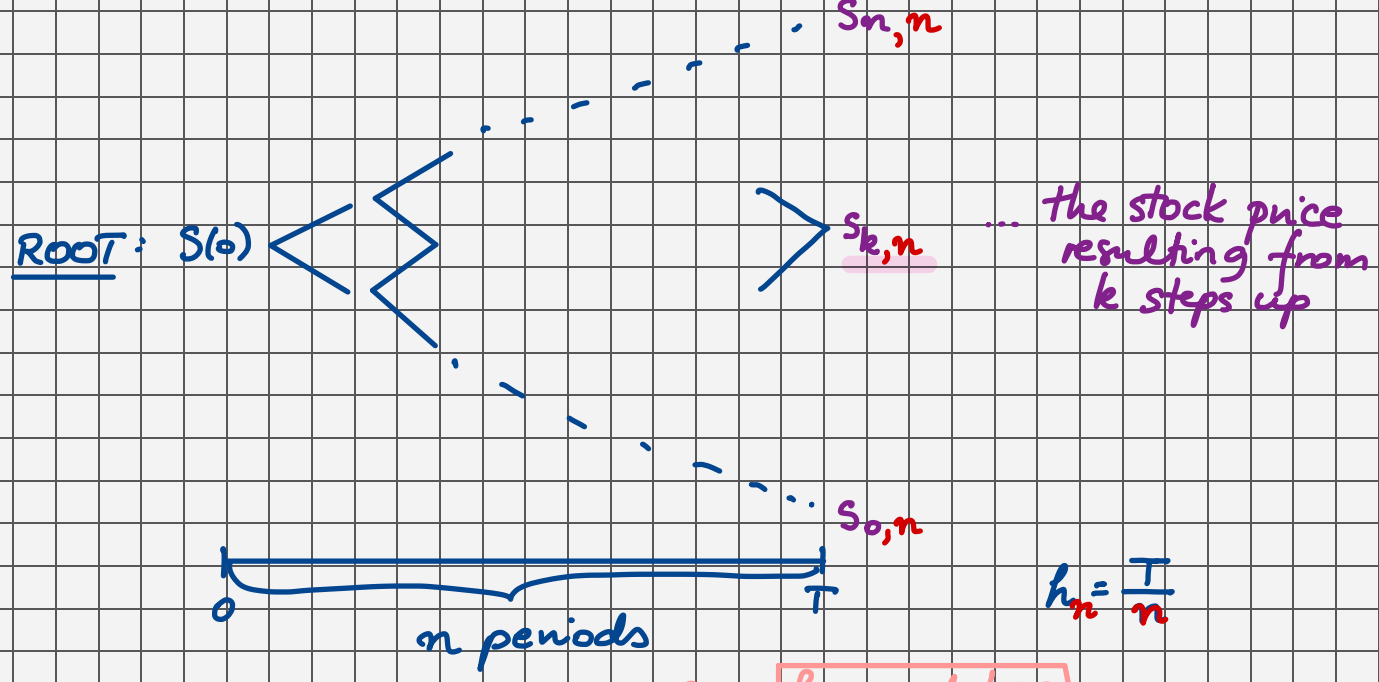
The n-Limit n-Period Binomial Tree.



$h_n = \frac{T}{n}$

u_n ... up factor

d_n ... down factor

e.g., forward tree

$$u_n = \exp\left(r \cdot \frac{T}{n} + \sigma\sqrt{\frac{T}{n}}\right)$$

$$d_n = \exp\left(r \cdot \frac{T}{n} - \sigma\sqrt{\frac{T}{n}}\right)$$

$$S_{k,n} = S(0) \cdot u_n^k \cdot d_n^{n-k} = S(0)\left(\frac{u_n}{d_n}\right)^k \cdot d_n^n$$

k ... corresponds to a realization of a binomial r.v.

w/ n trials

and

success probability

$$p_n^* = \frac{e^{r \cdot \frac{T}{n}} - d_n}{u_n - d_n}$$

e.g., in the forward tree

$$p_n^* = \frac{1}{1 + e^{\sigma\sqrt{\frac{T}{n}}}} \xrightarrow[n \to \infty]{} \frac{1}{2}$$

X_n ... # of steps up in n periods

$X_n \sim$ Binomial (# of trials $= n$, success prob. $= p_n^*$)

Q: Can we simply use the normal approximation to the binomial?

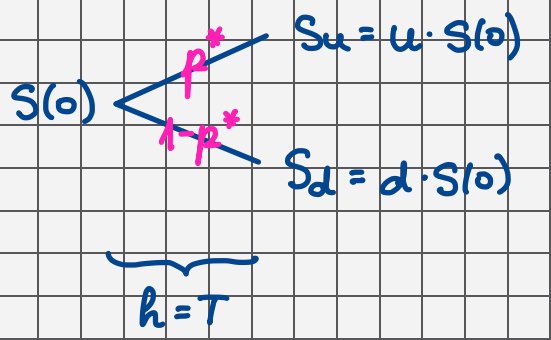
$Y_n \sim$ Binomial (n = # of trials, success probab. = p)

$\therefore$ p_n^* depends on n

but not all is lost

Subjective Probability.

Recall: When pricing, we use the risk-neutral probability measure $\mathbb{P}^*$ associated w/ $p^* = \frac{e^{rh} - d}{u - d}$



If we invest in one share of this non-dividend paying stock @ time-0, then the expected wealth @ time-T under $\mathbb{P}^*$ is

$$\mathbb{E}^*[S(T)] = S(0)e^{rT}$$

In Contrast:

There can be a subjective probability measure $\mathbb{P}$ associated w/ some probability of an upstep p.

We can think about the quality of our investment under $\mathbb{P}$, i.e., $\mathbb{E}[S(T)] = S(0)e^{\alpha \cdot T}$.

We call this α the mean rate of return.

In a binomial tree, we can express the "true" probability of a step up as

$$p = \frac{e^{\alpha h} - d}{u - d}$$

The mean rate of return on the stock under $\mathbb{P}^*$ is r

Moment Generating Functions.

For a r.v. Y, and for an independent argument denoted by t we define the moment generating f'tion (mgf) of Y as this function of t

$$M_Y(t) := \mathbb{E}\left[e^{Y \cdot t}\right]$$

for all t such that the expectation exists

Q: $M_Y(0) = 1$ => @ least $t=0$ is in the domain

We say that the mgf exists if it is finite for t such that $|t| \le b$ for some $b > 0$.

$\nu = nu$

Goal: To understand e^X w/ $X \sim \text{Normal}(\text{mean} = m, \text{var} = \nu^2)$

Recall: If $Z \sim N(0,1)$, $X = m + \nu \cdot Z$

Fact: $M_Z(t) = e^{\frac{t^2}{2}}$ for all $t \in \mathbb{R}$

=> For any normal X

$$M_X(t) = \mathbb{E}\left[e^{Xt}\right] = \mathbb{E}\left[e^{(m+\nu \cdot Z)\cdot t}\right]$$

$$= e^{mt}\,\mathbb{E}\left[e^{\nu t Z}\right]$$

$$= e^{mt} \cdot M_Z(\nu t) = e^{mt} \cdot e^{\frac{\nu^2 t^2}{2}} = e^{m \cdot t + \frac{\nu^2 \cdot t^2}{2}}$$